

# **USB Port Intrusion Detection and Access Control System Using Arduino**

## **Abstract**

The USB Port Intrusion Detection and Access Control System Using Arduino is designed to enhance computer and data security by preventing unauthorized access through removable USB devices. The proposed system continuously monitors USB ports and verifies whether a connected device is authorized before granting access to the host system. By implementing an access control mechanism based on device authentication, the system helps prevent data theft, malware infections, and unauthorized file transfers commonly associated with external storage devices.

In addition to access control, the system incorporates an intrusion detection module that identifies suspicious or unrecognized USB devices and immediately triggers security responses, such as visual alerts, audible alarms, or event logging. The system can also be configured to block unauthorized devices automatically, ensuring that sensitive information remains protected from external threats.

Built around the Arduino platform, the proposed solution provides a low-cost, reliable, and easy-to-deploy security mechanism suitable for educational institutions, offices, research laboratories, and organizations handling confidential information. By combining real-time USB monitoring, device authentication, and intrusion detection capabilities, the system significantly strengthens endpoint security and reduces the risk of unauthorized physical access to digital resources.

# **Secure Electronic Voting Machine with Voter Authentication, Encrypted Vote Storage, and Tamper Detection Using Arduino**

## **Abstract**

The Secure Electronic Voting Machine (EVM) with Voter Authentication, Encrypted Vote Storage, and Tamper Detection Using Arduino is designed to provide a secure, transparent, and trustworthy voting process by addressing the security challenges associated with conventional electronic voting systems. The proposed system integrates a robust voter authentication mechanism using technologies such as RFID cards or fingerprint recognition to ensure that only authorized voters can participate and that each voter is allowed to cast a vote only once.

To preserve voter privacy and maintain the confidentiality of election data, all votes are securely stored using encryption techniques, protecting them from unauthorized access, modification, or manipulation. The system further incorporates a tamper detection module that continuously monitors the voting unit for unauthorized physical access or interference attempts. In the event of tampering, the system immediately triggers alerts and records the incident to maintain the integrity of the election process.

Built around the Arduino platform, the proposed solution offers a low-cost, reliable, and scalable approach for conducting secure elections in educational institutions, organizations, and small-scale public voting environments. By combining voter authentication, encrypted vote management, and real-time tamper detection, the system significantly enhances election security, accuracy, and public confidence in electronic voting technologies.

# **Secure Wireless Communication System for IoT Devices Using AES Encryption and Intrusion Detection**

## **Abstract**

The Secure Wireless Communication System for IoT Devices Using AES Encryption and Intrusion Detection is designed to address security challenges in Internet of Things (IoT) environments. As IoT devices increasingly rely on wireless communication, protecting data from unauthorized access and cyberattacks has become essential. The proposed system uses the Advanced Encryption Standard (AES) algorithm to encrypt data before transmission, ensuring confidentiality and secure communication between connected devices.

To further strengthen security, the system incorporates an intrusion detection mechanism that monitors network activity and identifies suspicious behavior such as unauthorized access attempts and abnormal communication patterns. When an intrusion is detected, the system generates alerts and initiates preventive actions to protect connected devices.

Built using the Arduino platform and wireless communication modules, the proposed solution provides a low-cost, scalable, and efficient approach for securing IoT ecosystems in applications such as smart homes, healthcare systems, industrial automation, and environmental monitoring.

